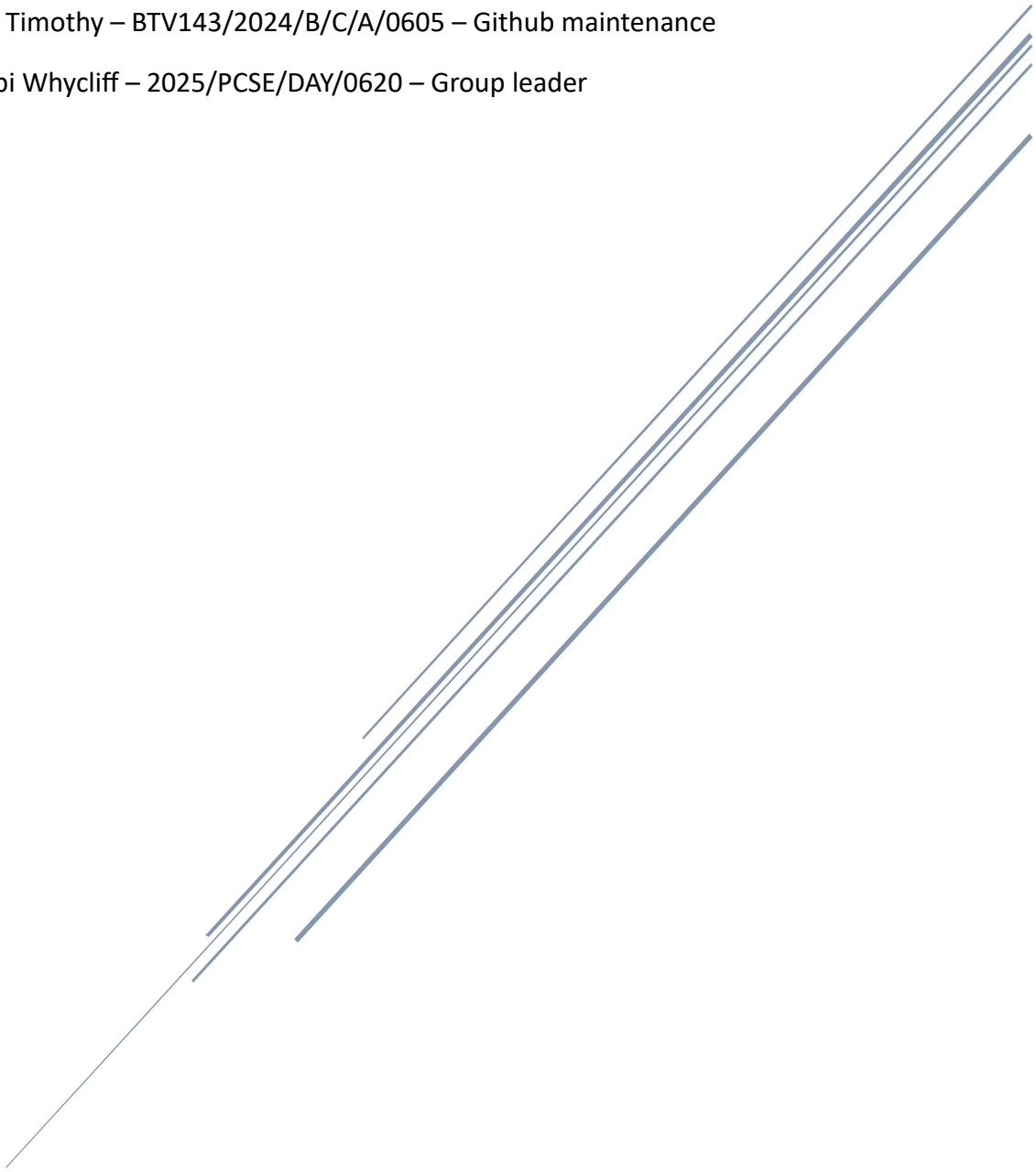
UGANDA INSTITUTE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY

GROUP PROJECT: ENGINEERING A SOLUTION FOR A REAL ORGANISATION

Group Members

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Software Engineering Principles
PCSE211

PCSE211 GROUP PROJECT — PART 1

Problem and Feasibility: Egg Production Tracking System

Brenda's Poultry Farm, Muyenga, Kampala

1. Problem statement

Brenda's Poultry Farm in Muyenga keeps chickens for egg production. The farm has about 70 laying hens and collects around 50–60 eggs a day. One worker records production in a large counter book, and Brenda, the owner, checks the entries and confirms the totals.

The main problem is that production totals are added by hand. This can lead to calculation mistakes and make it hard to know the correct number of eggs produced each day or week. It can also make it difficult to compare production over time. The worker who records the eggs and Brenda, who checks the figures, are both affected.

The group proposes a small smartphone-based Egg Production Tracking System. A worker will enter the daily number of eggs collected and record damaged eggs separately. The system will calculate daily and weekly totals automatically. It will focus only on egg production records, not sales, general inventory or flock health. This project is about analysis and design; no coding is required.

2. Chosen development model: Iterative model

The group will use the iterative model. This means planning a simple version, showing it to the lecturer, getting feedback, and improving the design in stages.

- The lecturer will check each part, so feedback can be used before moving to the next stage.
- The system is small, so its screens and functions can be reviewed and changed without redesigning everything.

3. Alternatives compared

Option	Advantages	Disadvantages
Keep using the counter book	No new equipment or training is needed. Staff already know it.	Totals still need to be added by hand, so mistakes can continue.
Use a spreadsheet on a phone	Can calculate totals automatically and may be low-cost.	A spreadsheet may be awkward to use on a small phone screen.
Small smartphone tracking system	Designed for the farm's egg records; calculates totals and keeps records together.	Needs setup, basic training and a way to back up records.

4. Feasibility study (TELOS)

Area	Assessment	Reason
Technical	Possible	A simple phone-based system can use entry forms and automatic totals. The farm needs a suitable smartphone.
Economic	Likely affordable	Free or low-cost tools may be used. Setup, training and backup costs should be checked.
Legal	Likely feasible	The system will mainly store egg production figures, and there is no obvious legal barrier to recording these digitally. If it stores workers' names or other personal details, the farm should protect that information, limit access to authorized users, and follow Uganda's Data Protection and Privacy Act, 2019.
Operational	Possible	A worker already records production and Brenda checks it. Short training and simple screens should help.
Schedule	Feasible	The group can prepare the design in stages and show each part to the lecturer for feedback.

5. Cost-benefit analysis and payback period3

The amounts below are estimates for planning, not prices confirmed by the farm. They assume the farm already has a suitable smartphone and that a low-cost tool can be used.

Estimated item	Cost (UGX)
Initial setup and configuration	150,000
Basic user training	50,000
Total initial cost	200,000
Backup/support per month	5,000

Possible benefits are less time spent adding figures, fewer calculation errors and easier access to production totals. For an example, if the system saves one hour a day for 26 days a month, and that time is valued at UGX 3,000 per hour, the time saved is worth UGX 78,000 per month. After UGX 5,000 monthly support, the estimated net benefit is UGX 73,000 per month.

Payback period = initial cost ÷ monthly net benefit = UGX 200,000 ÷ UGX 73,000 = about 2.7 months. This is only an estimate based on assumptions. The group should check whether the time saved and costs are reasonable.

6. Three main risks

Risk	How to reduce it
Wrong figures entered	Keep the form simple and let Brenda check the totals regularly.
Worker finds the system difficult	Use clear labels and give the worker a short practice session.

Records are lost or phone is unavailable	Back up records regularly and keep the counter book as a temporary backup during changeover.

7. Recommendation

The group recommends planning a small smartphone-based Egg Production Tracking System. Compared with the counter book, it can calculate daily and weekly totals automatically and help reduce manual addition errors. The iterative model will let the group improve the design after feedback from the lecturer and farm users. Before use, the farm should confirm the estimated costs, smartphone availability and backup method.

PCSE211 GROUP PROJECT — PART 2

SOFTWARE REQUIREMENTS SPECIFICATION

Brenda's Poultry Farm — Egg Production Tracking System

Document	*Details*
Version	1.0
Date	20 October 2026
Prepared by	Group A Members
Client / organisation	Brenda's Poultry Farm, Muyenga

Version history

Version	*Date*	*Change*	*By*
1.0	20 October 2026	First SRS draft based on interview and observation notes.	Project group

1. Introduction

1.1 Purpose

This document describes the requirements for a small Egg Production Tracking System for Brenda's Poultry Farm in Muyenga. It will guide the project group when planning and designing the system and help the farm check whether it meets its main need.

1.2 Scope

The smartphone-only system will record egg production and reduce mistakes caused by manual calculations in the counter book.

In scope	Out of scope in this version
Record date and number of eggs collected.	Record sales or payments.
Record damaged eggs separately.	Manage general stock or inventory.
Calculate daily and weekly production totals.	Manage flock health or treatment.
View saved production records and summaries.	Provide a desktop/computer version.

1.3 Definitions

Term	Meaning
Production record	Eggs collected and recorded for a particular date.
Damaged eggs	Eggs with cracks or other damage, recorded separately.
Daily total	Number of eggs recorded for one day.
Weekly total	Eggs recorded over a seven-day period.

1.4 References

Questionnaire interview notes for Brenda’s Poultry Farm (19 October 2026); observation of two egg-production counter books (19 October 2026); PCSE211 Software Engineering Principles, Unit 4, Section 4.8 sample SRS.

2. Overall description

2.1 Product perspective

This proposed system will support the farm’s current paper-based records. It will save egg-production entries and calculate totals automatically. It will not manage other farm activities.

2.2 Users

USER	MAIN TASKS	EXPECTED SKILLS	APPROXIMATE NUMBER
POULTRY WORKERS	Collect eggs from the hens' houses, sort eggs, identify damaged eggs and count the eggs collected each day.	Basic poultry farming and counting skills	3
BRENDA (OWNER)	Review production records, check the daily and weekly totals, and confirm that the figures are correct.	Basic smartphone skills	1
RECORD-KEEPING WORKER	Enter daily egg production figures into the smartphone system and check that the entries are correct.	Basic smartphone skills	1

2.3 Operating environment

The system will be used on a smartphone. Screens should fit the phone display and use clear labels and simple forms.

2.4 Constraints

The first version must stay small and focus on egg production tracking. The farm has about 70 laying hens and produces around 50–60 eggs per day. No computer version is planned.

2.5 Assumptions and dependencies

A worker will enter daily figures and Brenda will check and confirm totals. The system depends on users entering records regularly and correctly. Technical details and backup arrangements will be decided during later design.

3. Priority guide: Must = essential to the main purpose; Should = important but not essential for the first usable version; Could = useful extra; Won't this time = excluded from this version.

ID	*Requirement*	*Priority*	*Source*
FR-01	The system shall let a worker enter the date and number of eggs collected each day.	Must	Interview
FR-02	The system shall let the user record damaged eggs separately from intact eggs.	Must	Interview
FR-03	The system shall save each production entry for later viewing.	Must	Observation; project scope
FR-04	The system shall calculate the total eggs recorded for a selected day.	Must	Interview: manual calculation errors
FR-05	The system shall calculate the total eggs recorded for a selected week.	Must	Interview: need for weekly figures
FR-06	The system shall show intact and damaged egg counts separately for a selected day.	Must	Interview
FR-07	The system shall let Brenda review and confirm saved production entries.	Should	Interview: owner checks totals
FR-08	The system shall let a user view previous production records by date.	Should	Observation; project scope
FR-09	The system shall warn the user if a required field, such as date or egg count, is missing.	Should	Project scope: reduce incomplete records
FR-10	The system shall let a user correct an entry when a recording mistake is found.	Should	Project scope: support accuracy

4. Non-functional requirements

The targets below are proposed and can be checked during testing.

ID	*Type*	*SMART requirement*	*Priority*
NFR-01	Performance	The system shall display a requested daily or weekly total within 3 seconds under normal conditions.	Must
NFR-02	Usability	A worker with basic smartphone skills shall enter and save a sample record correctly after no more than 30 minutes of training.	Must
NFR-03	Security	The system shall require an authorised user to sign in before viewing or changing saved records.	Must
NFR-04	Reliability	A saved production record shall still be available after the user closes and reopens the system.	Must
NFR-05	Compatibility	The main entry and summary screens shall fit a smartphone display without horizontal scrolling.	Should

5. External interface

User interface: a smartphone screen with simple fields for date, intact egg count and damaged egg count, plus clear buttons to save entries and view totals. The system should use plain English and readable text.

6. Acceptance criteria

Requirement	*Test*	*Expected result*
FR-02, FR-06	Enter 55 intact eggs and 3 damaged eggs for one date.	The two counts remain separate and display correctly.
FR-04, FR-05	Enter sample records for several days and request daily and weekly totals.	Totals match the sum of saved entries.
NFR-01	Request a daily or weekly total and time the result.	The total appears within 3 seconds.
NFR-04	Save a record, close the system and reopen it.	The saved record is still available.

7. MoSCoW and SMART justification

Recording production, separating damaged eggs, saving records and calculating totals are Must-have requirements because without them the system would not solve the main problem. Reviewing past records and correcting entries are Should-have requirements because they improve accuracy but come after the basic functions. Sales and flock-health features are excluded to keep this version focused. The non-functional requirements are measurable: for example, the total must appear within 3 seconds and a worker must be able to complete a sample entry after no more than 30 minutes of training.

8. Requirements-gathering evidence

Interview: the worker explained that daily production is written in a large counter book, manual calculations can be wrong, damaged eggs are counted separately, and Brenda checks and confirms totals. The farm has about 70 laying hens and produces around 50–60 eggs per day.

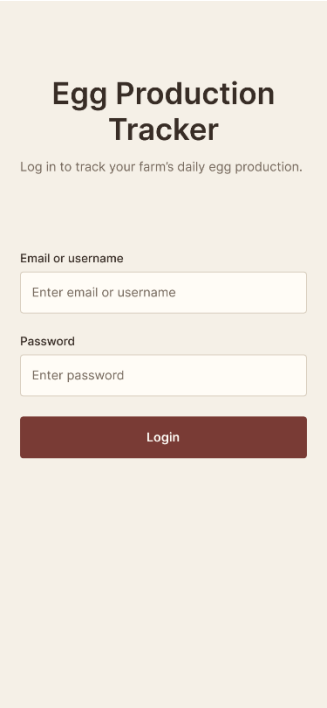
Observation: about two counter books containing egg-production records were observed. This supports the need for an easier way to save and find production figures. Any extra details about the books should only be added if they were actually seen.

PCSE211 GROUP PROJECT — PART 3

Analysis and design

Brenda's Poultry Farm — Egg Production Tracking System

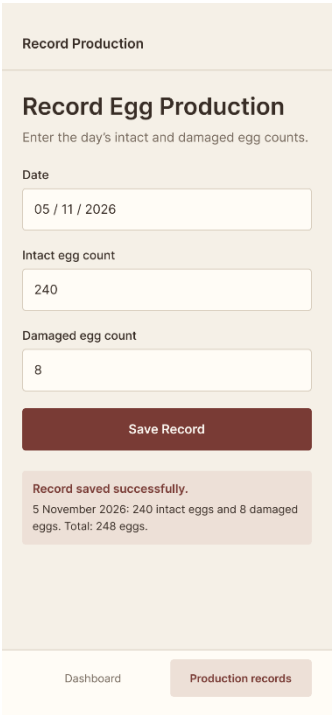
UI Screens



Login Screen

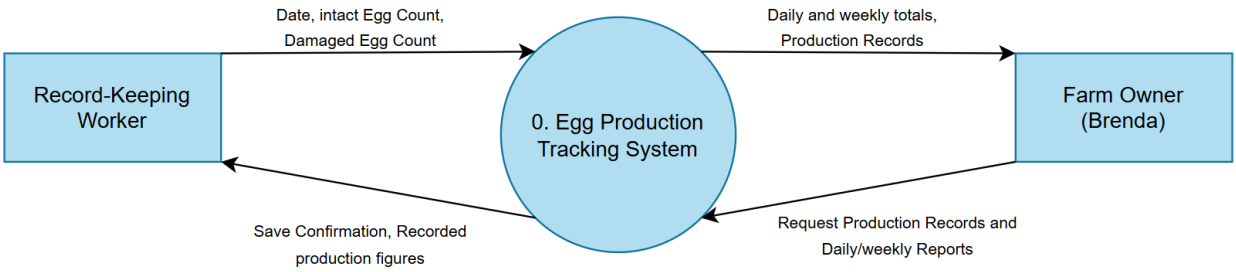


Dashboard Screen

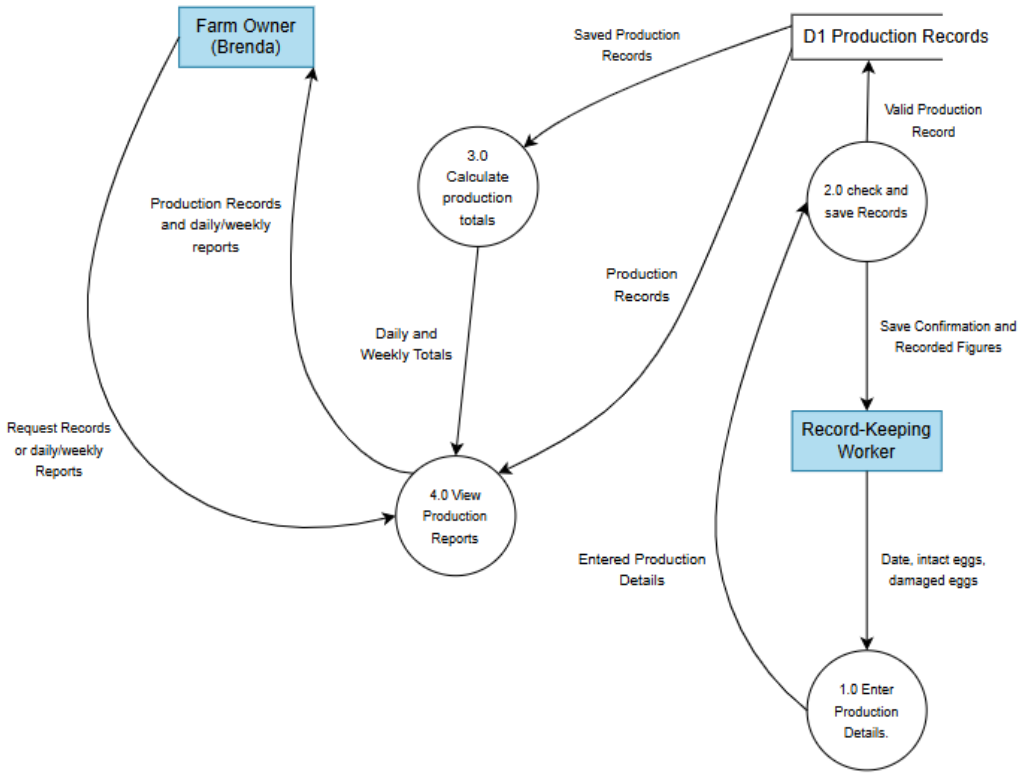


Recording Screen

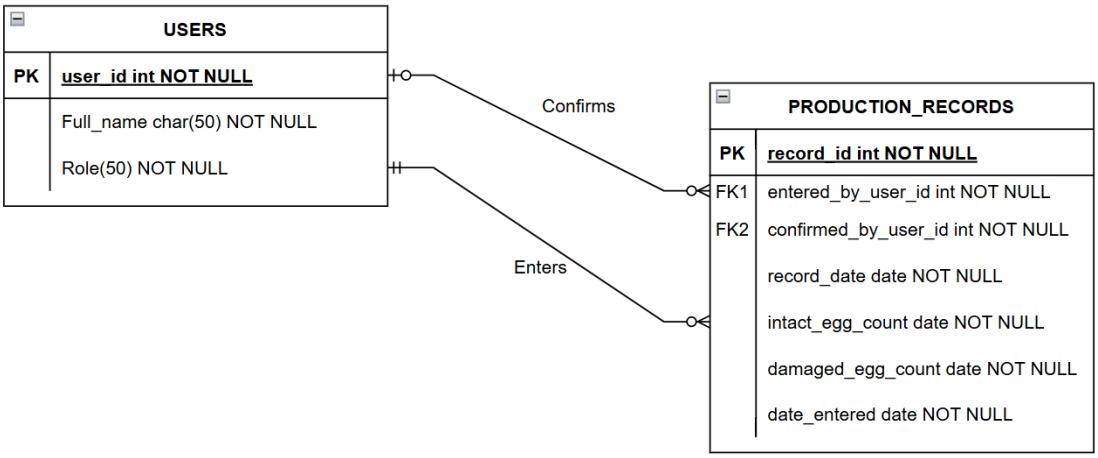
Context Diagram



Level-1 DFD



ERD



DATA DICTIONARY

Egg Production Tracking System
Brenda's Poultry Farm, Muyenga

This data dictionary describes the fields used in the Entity Relationship Diagram (ERD). It explains the information stored in each entity, its data type, key and meaning.

1. USER

This entity stores information about the people who use the system.

Field name	Data type	Key	Description
user_id	INT	PK	Unique number given to each user.
full_name	VARCHAR(100)	—	Full name of the user.
role	VARCHAR(30)	—	Shows whether the user is the Farm Owner or Record-Keeping Worker.

2. PRODUCTION_RECORD

This entity stores the daily egg production information entered into the system.

Field name	Data type	Key	Description
record_id	INT	PK	Unique number given to each production record.
record_date	DATE	—	Date when the eggs were collected.
intact_egg_count	INT	—	Number of intact eggs collected.
damaged_egg_count	INT	—	Number of damaged or broken eggs collected.
date_entered	DATETIME	—	Date and time the record was entered into the system.
entered_by_user_id	INT	FK	Identifies the user who entered the production record.

3. Notes

Term	Meaning
INT	Used for whole numbers, such as egg counts and identification numbers.
VARCHAR	Used for text, such as names and user roles.
DATE	Stores a date.
DATETIME	Stores both date and time.
PK (Primary Key)	Uniquely identifies each record.
FK (Foreign Key)	Connects one table to another.

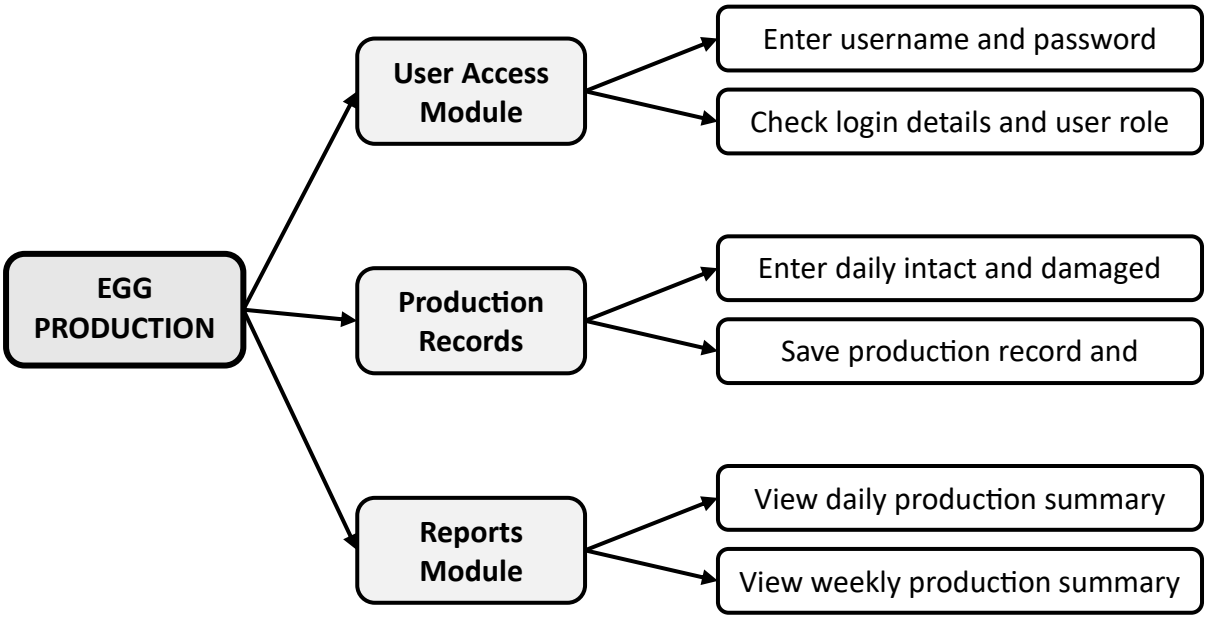
Calculated total: The system can calculate the total eggs collected by adding the intact eggs and damaged eggs. This total does not need to be stored as a separate field because it can be calculated when needed.

Total eggs = Intact eggs + Damaged eggs

Note: The current design does not include a field for Brenda's confirmation because confirmation inside the smartphone system has not been included in the agreed requirements.

Model Diagram

The diagram below shows how the system is divided into smaller modules. Each module is responsible for a main task in recording and viewing egg production.



Explanation of the Modules

1. User Access Module

This module allows the farm owner and the record-keeping worker to access the system using their login details. It checks the username and password and identifies the user's role. This helps prevent people who are not allowed to use the system from accessing the records.

2. Production Records Module

This is the main module of the system. It allows the worker to enter the number of intact eggs and damaged eggs collected each day. It saves the information as a production record and calculates the total eggs collected. The damaged eggs remain recorded separately, as the farm currently handles them separately.

3. Reports Module

This module uses the saved production records to show daily and weekly egg production summaries. It helps the owner check production figures over time without having to calculate totals manually from the counter book.

Cohesion

Cohesion means how well the tasks inside one module belong together. The system aims for functional cohesion, which is the best type shown in the lecturer's notes. This means each module has one clear job.

In the diagram, each module in the middle column is connected only to the tasks it performs on the right, and every task sits under one module only. The User Access Module is connected to "Enter username and password" and "Check login details and user role", and both tasks serve one purpose, which is letting the right person log in. The Production Records Module is connected to "Enter daily intact and damaged egg counts" and "Save production record and calculate total eggs", and both tasks are about capturing the day's egg production. The Reports Module is connected to "View daily production summary" and "View weekly production summary", and both tasks are about displaying production summaries. No task in the diagram looks out of place in its module, so the tasks in each module support its main job.

Coupling

Coupling means how much one module depends on another module. The system aims for data coupling, which is the best type shown in the lecturer's notes. This means modules share only the information they need.

In the diagram, the User Access, Production Records and Reports modules each branch separately from the Egg Production Tracking System box, and there are no arrows running directly between the modules. This shows that none of them controls or changes the inner workings of another. The only link between them is data: the Reports Module uses the saved production records to prepare the daily and weekly summaries shown on the right of the diagram. It does not need to control how the Production Records Module works or change its internal data directly. Keeping the modules separate in this way makes the system easier to understand and update.

Overall, the system is divided into modules with clear responsibilities. This gives each module good functional cohesion and keeps the connection between modules simple through data coupling.

PCSE211 GROUP PROJECT — PART 4

Managing and maintaining the system

Brenda's Poultry Farm — Egg Production Tracking System

Github Repository: here

Change Request and Software Maintenance Plan

Handling a Change Request

If the farm owner or a worker requests a change to the Egg Production Tracking System, the group will first record and understand the request. The group will then discuss why the change is needed and how it may affect the system. The requested change will be checked against the project scope, time and available resources. If approved, the group will update the requirements and relevant designs, make the change, and test it to ensure that the system still works correctly. The updated version will then be reviewed and documented.

Types of Software Maintenance

1. **Corrective maintenance:** Fixing an error where the system calculates the daily egg production total incorrectly.
2. **Adaptive maintenance:** Updating the system so that it continues to work when the farm changes to a newer smartphone or operating system.
3. **Perfective maintenance:** Improving the production reports so Brenda can view weekly egg production more clearly.
4. **Preventive maintenance:** Regularly backing up production records and reviewing the system to identify possible problems before they cause failures.

APPENDICES

Appendix A: Interview notes The following two pages reproduce the interview notes supplied by the project group. The respondent is identified as a farm worker whose name was not recorded. The source document itself notes that the draft should be checked against what the worker actually said before submission.

INTERVIEW NOTES — BRENDA'S POULTRY FARM

PCSE211 Group Project: Egg Production Tracking System

Organization	Brenda's Poultry Farm
Location	Muyenga, Kampala, Uganda

Date and method	19 October 2026 — Questionnaire
Respondent	Farm worker (name not recorded)

Note: This draft is based on information provided by the project group. Please check that the details match what the worker said before submitting it.

1. Purpose of the questionnaire

The questionnaire was used to learn how the farm keeps egg production records, what problems workers face when using the records, and what information the farm needs. The answers will help the group plan a simple Egg Production Tracking System.

2. Questions and responses

Question	Response
1. What does the farm mainly do?	The farm keeps chickens for egg production.
2. About how many laying hens does the farm have?	The farm has about 70 laying hens.
3. How many eggs does the farm collect?	The farm collects about 50 to 60 eggs a day. This is around 350 to 420 eggs in a week.
4. Who collects and records the eggs?	Farm workers collect the eggs and record the production.
5. How do you record the eggs collected each day?	The farm uses a large counter book meant for recording egg production.
6. What is the main problem with the current record book?	Workers have to calculate the totals by hand. Sometimes they make mistakes when adding up the figures.
7. How do these mistakes affect the farm?	When figures are added wrongly, it becomes hard to know the correct production total and compare production over time.
8. Have you had problems with missing or wrong records?	The problem reported for this project is mistakes during manual calculations. Whether records have also gone missing should be checked if needed.
9. How do you work out weekly or monthly production?	Production figures are taken from the counter book and added by hand, so calculation mistakes can happen.
10. What happens to eggs that are cracked or damaged?	If an egg is cracked but the inner membrane is still intact, the family may eat it at home or sell it cheaply to local people for immediate use. If the shell is badly cracked, the egg is thrown away.
11. What reports would be useful to the farm?	Daily and weekly egg production reports are useful. Flock health, sales and inventory reports were also mentioned, but this project will focus on egg production records.
12. Would a simple computerised system help? Why?	A system could add up production figures automatically and make daily and weekly totals easier to check. This benefit should be confirmed with the worker.
13. What should the new system be able to do?	It should record the number of eggs collected each day and show daily and weekly totals. It could also record damaged eggs separately if needed.

14. Is there anything else we should know about the current process?	The main point for the project is that production is written in a counter book and totals are calculated by hand. Any other details should be checked with the farm worker.

3. Main findings

- The farm has about 70 laying hens and collects around 50–60 eggs per day, or about 350–420 eggs per week.
- Egg production is recorded in a large counter book.
- Workers add up the figures by hand. Mistakes can make it hard to track production correctly over time.
- Cracked eggs with the inner membrane still intact may be eaten at home or sold cheaply for immediate use. Eggs with badly cracked shells are thrown away.

4. Proposed system and scope

The proposed system will focus on egg production records only. It should let a worker enter the number of eggs collected each day and calculate daily and weekly totals automatically. A separate count of damaged eggs can be included if the farm needs it. Sales management, general inventory and flock health will not be separate functions because the project is meant to solve one main problem.